

TrES-3b

R-1900

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140523 · created 2026-08-03T14:06:13+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456800.895 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.83 +/- 2.24 [%]
Orbital Inclination (inc)	86.37 +/- 2.94
Airmass coefficient 1 (a1)	2.14 +/- 0.2
Airmass coefficient 2 (a2)	-0.71 +/- 0.92
Scatter in the residuals of the lightcurve fit is	9.82 %
Best Comparison Star	#2 - [543, 167]
Optimal Method	PSF photometry
Transit Duration (day)	0.078 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.196 ± 0.057 vs 0.16309 (deviation 0.58σ)
Duration fit vs published	0.078 d vs 0.0567 d (deviation 1.64σ)
Mid-time O-C	-2.3 min
Depth SNR	3.4
Residual scatter	9.82 %

Light curve

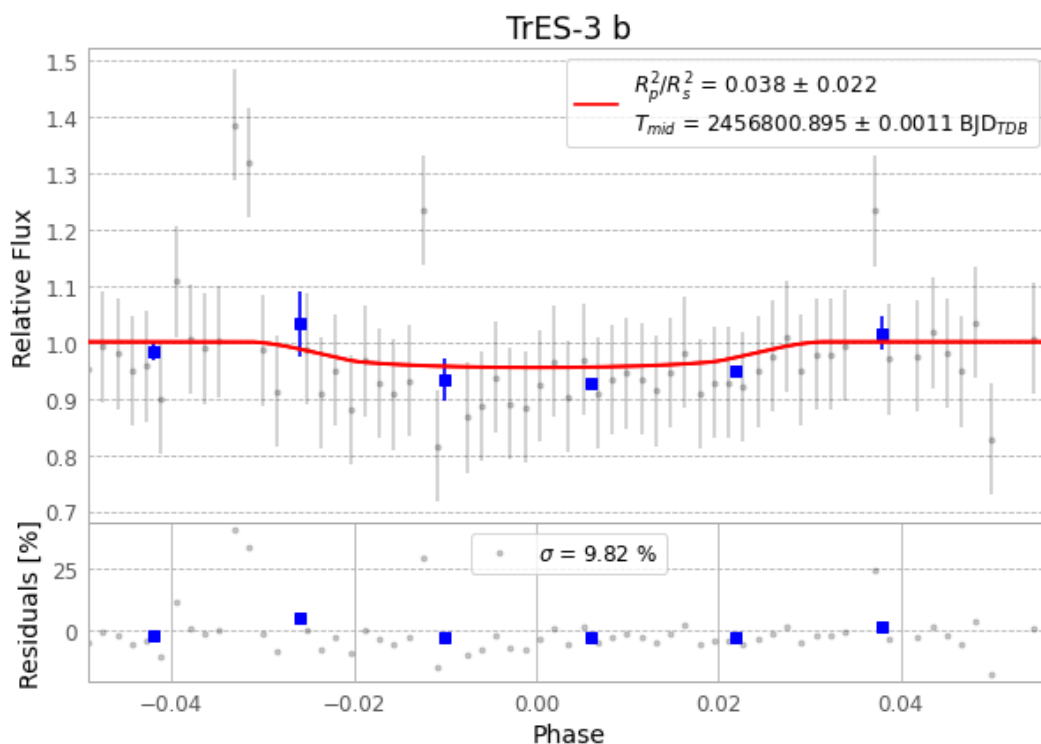


Figure 1 — FinalLightCurve_TrES-3 b_2014-05-23.png (SHA-256 0a0796248a53d367...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 277], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d5b90d038b2b2403...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3140523075112.FITS → TRES-3140523110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140523.log (31d073a9853c...), FinalLightCurve_TrES-3 b_2014-05-23.png (0a0796248a53...), FinalLightCurve_TrES-3 b_2014-05-23.csv (36edf5711574...)

Compute resources

Wall time 350.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 14:06Z.